Implement Page Rank Algorithm. (Use python or beautiful soup for implementation).

PageRank is an algorithm developed by Google to rank web pages in search results based on their importance.

- It models the web as a directed graph:
- Each webpage is a node.
- Each hyperlink is a directed edge from one node to another.

✅ 📊 **PageRank Formula**

For a page P, its PageRank is computed as:

$$PR(P) = \frac{1-d}{N} + d \sum_{Q \in In(P)} \frac{PR(Q)}{L(Q)}$$

Where:
PR(P)PR(P)PR(P) = PageRank of page P
ddd = Damping factor (usually 0.85)
NNN = Total number of pages
In(P)In(P)In(P) = Set of pages that link to page P
L(Q)L(Q)L(Q) = Number of outbound links from page Q

Code :

```
import requests
from bs4 import BeautifulSoup
from urllib.parse import urljoin, urlparse
import numpy as np
import time

MAX_PAGES = 10
MAX_DEPTH = 2

def get_links(url):
    try:
        soup = BeautifulSoup(requests.get(url, timeout=5).content, 'html.parser')
        links = {urljoin(url, a['href']).split('#')[0].rstrip('/')
                 for a in soup.find_all('a', href=True)
                 if urlparse(a['href']).scheme in ('http', 'https') or a['href'].startswith('/')}
        return links
    except:
        return set()
```

```
def crawl(seed_url):
    visited, graph, queue = set(), {}, [(seed_url, 0)]
    while queue and len(visited) < MAX_PAGES:
        url, depth = queue.pop(0)
        if url in visited or depth > MAX_DEPTH:
            continue
        print(f"Crawling: {url}")
        visited.add(url)
        links = get_links(url) - {url}
        graph[url] = links
        queue.extend((link, depth + 1) for link in links if link not in visited)
        time.sleep(1)
    return graph

def pagerank(graph, d=0.85, max_iter=100, tol=1e-6):
    nodes = list(graph.keys())
    n = len(nodes)
    ranks = np.ones(n) / n
    idx = {url: i for i, url in enumerate(nodes)}
    M = np.zeros((n, n))

    for i, url in enumerate(nodes):
        out_links = graph[url]
        if not out_links:
            M[:, i] = 1 / n
        else:
            for link in out_links:
                if link in idx:
                    M[idx[link], i] = 1 / len(out_links)

    for _ in range(max_iter):
        new_ranks = (1 - d) / n + d * M @ ranks
        if np.linalg.norm(new_ranks - ranks, 1) < tol:
            break
        ranks = new_ranks

    return dict(zip(nodes, ranks))

if __name__ == '__main__':
    seed = 'https://w3schools.com'
    graph = crawl(seed)
```

```
print("\nGraph:")
for url, links in graph.items():
    print(f"{url} -> {len(links)} links")

print("\nPageRank Scores:")
scores = pagerank(graph)
for url, score in sorted(scores.items(), key=lambda x: -x[1]):
    print(f"{url}: {score:.4f}")
```

Output :

```
Crawling: https://w3schools.com
Crawling: https://w3schools.com/about/default.asp
Crawling: https://w3schools.com/html/tryit.asp?filename=tryhtml_default_default
Crawling: https://w3schools.com/vue/index.php
Crawling: https://w3schools.com/js/tryit.asp?filename=tryjs_default
Crawling: https://w3schools.com/bootstrap/bootstrap_quiz.asp
Crawling: https://campus.w3schools.com/products/learn-typescript
Crawling: https://w3schools.com/go/go_exercises.php
Crawling: https://w3schools.com/cpp/default.asp
Crawling: https://w3schools.com/gen_ai/index.php

Graph:
https://w3schools.com -> 277 links
https://w3schools.com/about/default.asp -> 270 links
https://w3schools.com/html/tryit.asp?filename=tryhtml_default_default -> 5 links
https://w3schools.com/vue/index.php -> 270 links
https://w3schools.com/js/tryit.asp?filename=tryjs_default -> 5 links
https://w3schools.com/bootstrap/bootstrap_quiz.asp -> 270 links
https://campus.w3schools.com/products/learn-typescript -> 28 links
https://w3schools.com/go/go_exercises.php -> 269 links
https://w3schools.com/cpp/default.asp -> 271 links
https://w3schools.com/gen_ai/index.php -> 270 links

PageRank Scores:
https://campus.w3schools.com/products/learn-typescript: 0.0153
https://w3schools.com/cpp/default.asp: 0.0153
https://w3schools.com/about/default.asp: 0.0153
https://w3schools.com/vue/index.php: 0.0153
https://w3schools.com/bootstrap/bootstrap_quiz.asp: 0.0153
https://w3schools.com/gen_ai/index.php: 0.0153
https://w3schools.com/go/go_exercises.php: 0.0153
https://w3schools.com/html/tryit.asp?filename=tryhtml_default_default: 0.0150
https://w3schools.com/js/tryit.asp?filename=tryjs_default: 0.0150
https://w3schools.com: 0.0150
```